

LUNGS AND BRONCHOPULMONARY SEGMENTS

High-yield anatomy notes with enlarged source-PDF diagrams and image explanations

Core identity

The lungs are the principal organs of respiration and occupy the right and left pleural cavities on either side of the mediastinum.

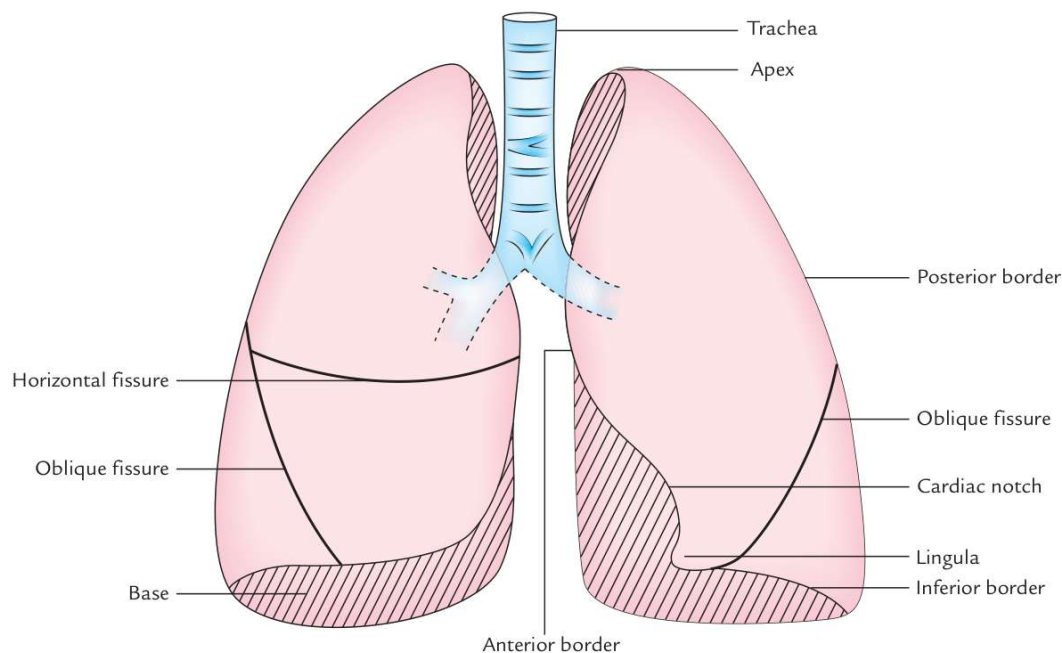
- Main role: exchange of oxygen and carbon dioxide between inspired air and blood.
- Shape: conical/pyramidal; apex extends into root of neck, base rests on diaphragm.
- Right lung: larger, shorter, broader and has three lobes.
- Left lung: smaller, longer, narrower and has two lobes with cardiac notch and lingula.

Why this topic matters

Lung anatomy is tested through external features, root/hilum arrangement, bronchial tree, surface markings, blood supply, nerve supply, lymph drainage and bronchopulmonary segments.

- Clinically important for auscultation, bronchoscopy, aspiration pneumonia, bronchiectasis and segmental lung resection.
- Bronchopulmonary segments are independent anatomical, functional and surgical units.

Fig. 18.1 - Trachea and lungs from front



Use this figure as the starting map. The right lung shows both oblique and horizontal fissures, while the left lung shows a cardiac notch and lingula. The inferior border forms the lower sharp margin and changes level during respiration.

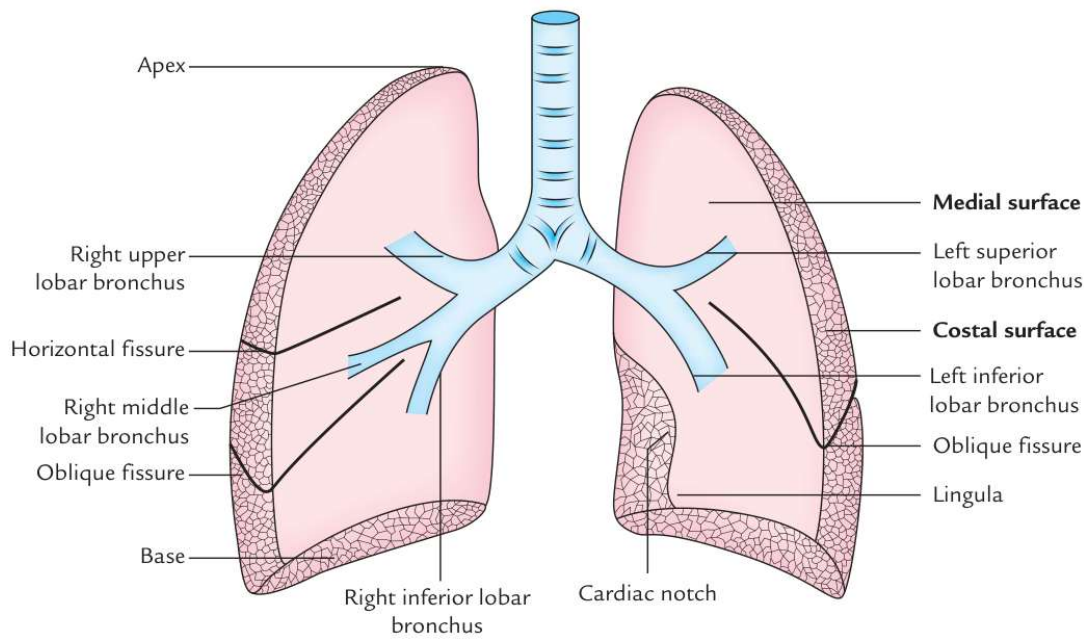
EXTERNAL FEATURES AND ANATOMICAL POSITION

External features of each lung

Each lung shows an apex, a base, three borders and two surfaces. These features are the framework for describing the lobes, fissures, root, hilum and relations.

- Apex: rounded upper end projecting above the first rib into the root of the neck.
- Base: concave diaphragmatic surface resting on the dome of the diaphragm.
- Borders: anterior, posterior and inferior.
- Surfaces: costal surface and medial surface.

Fig. 18.2 - Lobes of lungs with lobar bronchi



The figure connects lobes with their bronchi. The right principal bronchus supplies three lobes through superior, middle and inferior lobar bronchi. The left principal bronchus supplies superior and inferior lobes; the lingula represents a tongue-shaped projection of the superior lobe.

Side determination

- Hold the apex upwards and the base downwards.
- Keep the convex costal surface laterally and the medial surface with hilum medially.
- The thin anterior border faces forwards; the thick rounded posterior border faces backwards.
- Do not depend only on number of fissures and lobes because anatomical variation can occur.

APEX, BASE AND BORDERS

Apex of lung

The apex is the rounded superior end of the lung. It extends into the root of the neck about 3 cm above the anterior end of the first rib and about 2.5 cm above the medial one-third of the clavicle.

- Coverings: cervical pleura and suprapleural membrane.
- Anterior relations: subclavian artery, internal

Base of lung

The base is the lower semilunar concave surface resting on the diaphragm; it is also called the diaphragmatic surface.

- Right base is separated from the liver by the right dome of diaphragm.
- Left base is separated from spleen and fundus of stomach by the left dome of diaphragm.

- thoracic artery and scalenus anterior.
- Posterior relations: neck of first rib, T1 ventral ramus, first posterior intercostal vessels and sympathetic chain.

- The right base is deeper because the right dome of diaphragm is raised by the liver.

Borders at a glance

- Anterior border is thin and short; the left anterior border has cardiac notch and lingula.
- Posterior border is thick and rounded; it extends from C7 vertebra to T10 vertebra.
- Inferior border is semilunar and separates costal and medial surfaces.

Clinical correlation - Pancoast syndrome

Cancer at the apex of lung may involve structures related to the posterior aspect of the apex. It can produce pain along the medial forearm and hand, wasting of small hand muscles due to T1 involvement, Horner syndrome due to sympathetic chain involvement, and erosion of the first rib. Spread to nearby veins, arteries or phrenic nerve may produce venous engorgement, diminished upper limb pulse or paralysis of one hemidiaphragm.

SURFACES AND MEDIASTINAL RELATIONS

Surfaces

- Costal surface: large, smooth, convex and related to the thoracic wall through costal pleura and endothoracic fascia.
- Medial surface: divided into a posterior vertebral part and an anterior mediastinal part.
- The mediastinal surface presents the hilum and shows impressions from the heart, great vessels, trachea, esophagus and nerves.

Right mediastinal surface	Left mediastinal surface
Right atrium	Left ventricle
Superior and inferior venae cavae	Ascending aorta
Azygos vein	Arch and descending thoracic aorta
Right brachiocephalic vein	Left subclavian and left common carotid arteries
Esophagus and trachea	Esophagus and thoracic duct
Right phrenic, right vagus, right sympathetic chain	Left phrenic, left vagus, left recurrent laryngeal nerve, left sympathetic chain

Fig. 18.3 - Right mediastinal surface

Fig. 18.4 - Left mediastinal surface

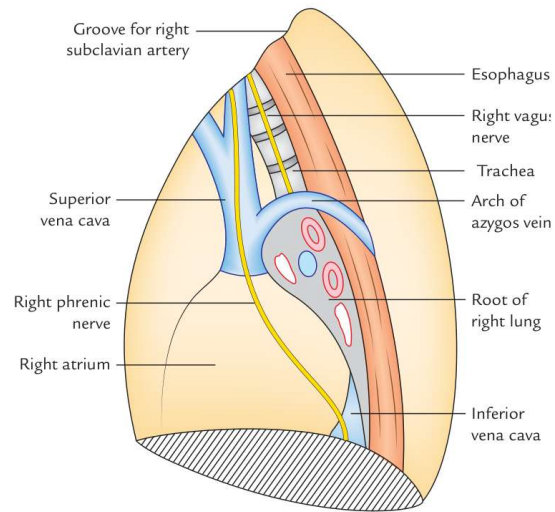
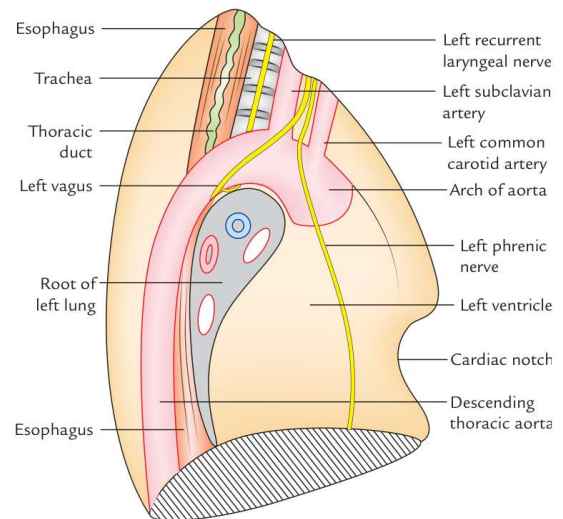


Fig. 18.3 Structures related to the mediastinal surface of the right lung.

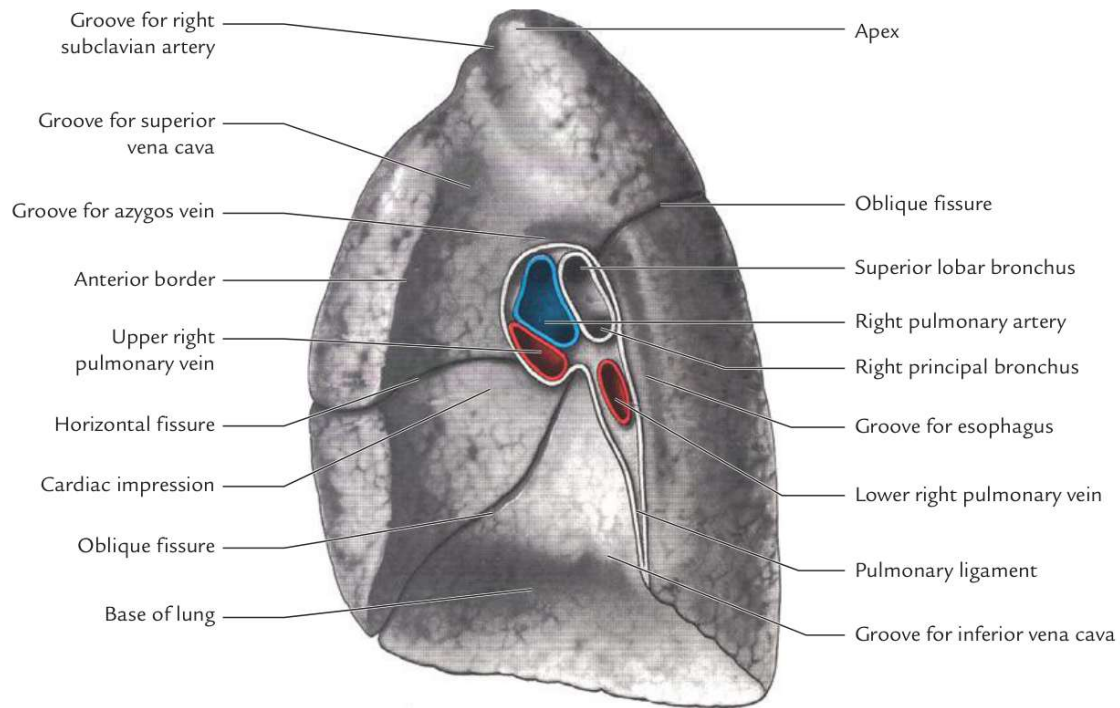
On the right side, the lung is closely related to SVC, right atrium, IVC, arch of azygos vein, esophagus, trachea and right phrenic/vagus nerves.



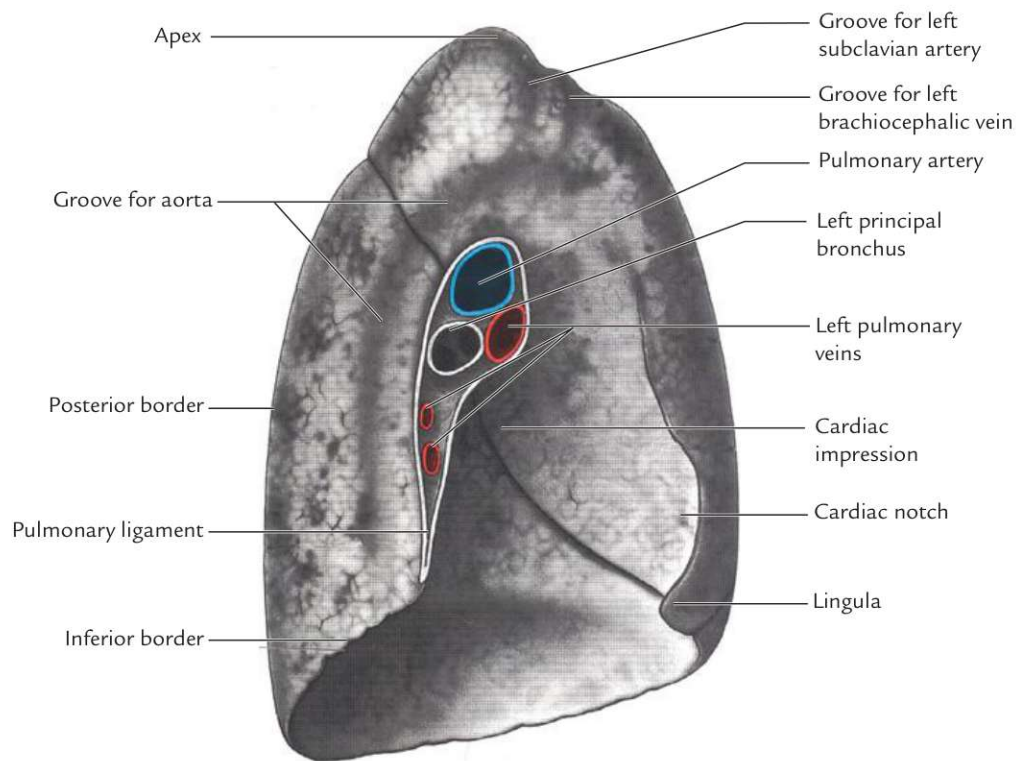
On the left side, the lung is related to the aortic arch, descending thoracic aorta, left ventricle, esophagus, thoracic duct and left recurrent laryngeal nerve.

How to read mediastinal-relation diagrams

- First identify the hilum/root: bronchus is posterior, pulmonary artery is superior/central, and pulmonary veins are anterior-inferior.
- On the right lung, venous structures dominate the medial surface: SVC, right atrium and IVC.
- On the left lung, arterial structures dominate: arch of aorta and descending thoracic aorta.
- Phrenic nerves pass anterior to the lung root; vagus nerves pass posterior to it.
- Use the cardiac notch and lingula to identify the left lung quickly in diagrams.

Fig. 18.5 - Medial impressions of right lung

The right lung has impressions for SVC, azygos arch, esophagus, right atrium, IVC and hilar structures. The blue outline marks pulmonary artery and the red outlines mark pulmonary veins at the hilum.

Fig. 18.6 - Medial impressions of left lung

The left lung carries prominent grooves for arch and descending aorta and the cardiac impression is deeper because of the left ventricle. The cardiac notch and lingula are characteristic left-lung features.

LOBES, FISSURES AND AUSCULTATION AREAS

Right lung: 3 lobes

The right lung is divided into superior, middle and inferior lobes by an oblique fissure and a horizontal fissure.

- Oblique fissure separates inferior lobe from superior and middle lobes.
- Horizontal fissure separates superior lobe from middle lobe.
- Common exam point: right lung is larger, shorter and broader.

Left lung: 2 lobes

The left lung is divided into superior and inferior lobes by one oblique fissure.

- Oblique fissure is more vertical than that of the right lung.
- Left lung has cardiac notch and lingula.
- Common exam point: left lung is smaller, longer and narrower because of the heart.

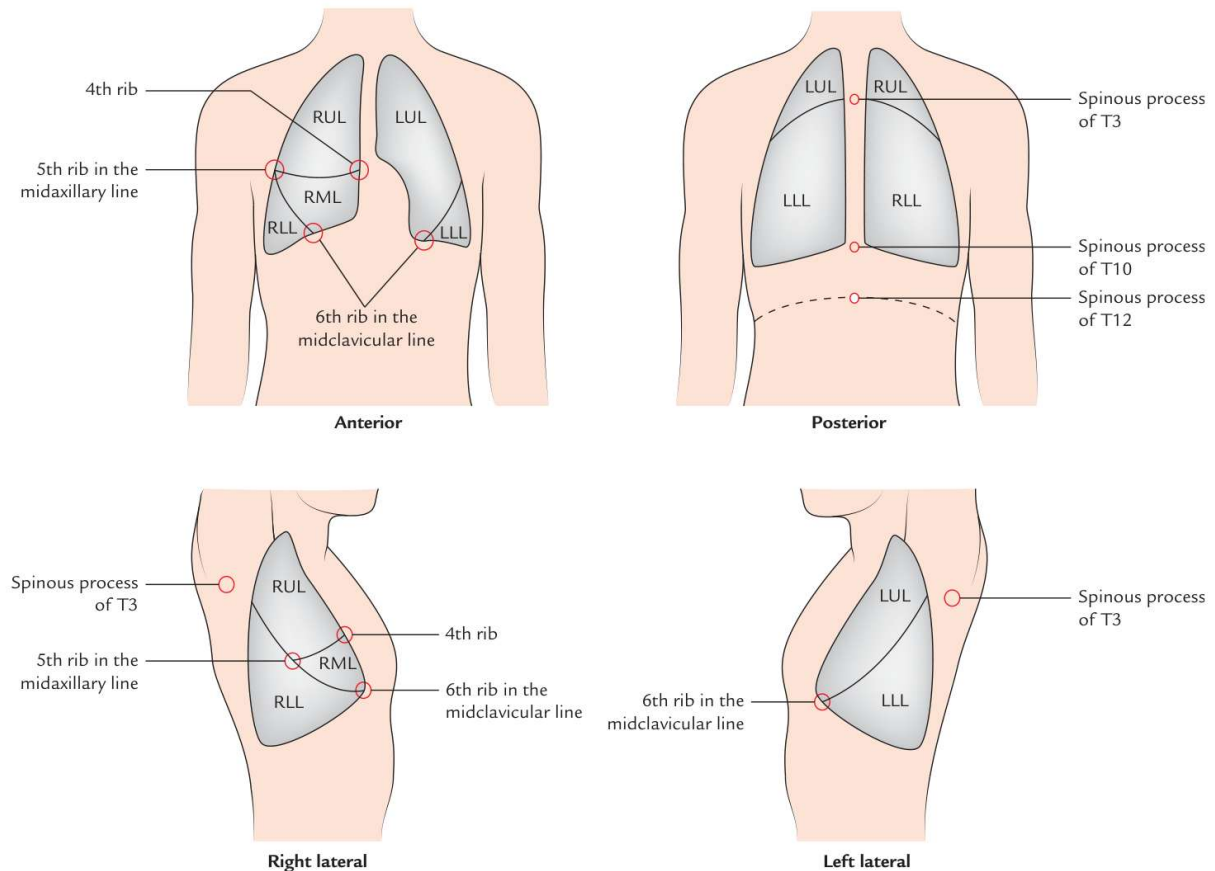
Fissure interpretation for exams

- The oblique fissure works like a plane of cleavage: upper parts expand forwards/laterally, lower parts expand downwards/backwards.
- Horizontal fissure belongs only to the right lung; a left horizontal fissure is a normal variant in some individuals.
- Incomplete fissures are clinically important because they may cause postoperative air leakage

after lobectomy.

- On X-ray, the horizontal fissure is commonly seen in PA view, while oblique fissures are better appreciated in lateral view.

Fig. 18.10 - Surface projection of lung lobes

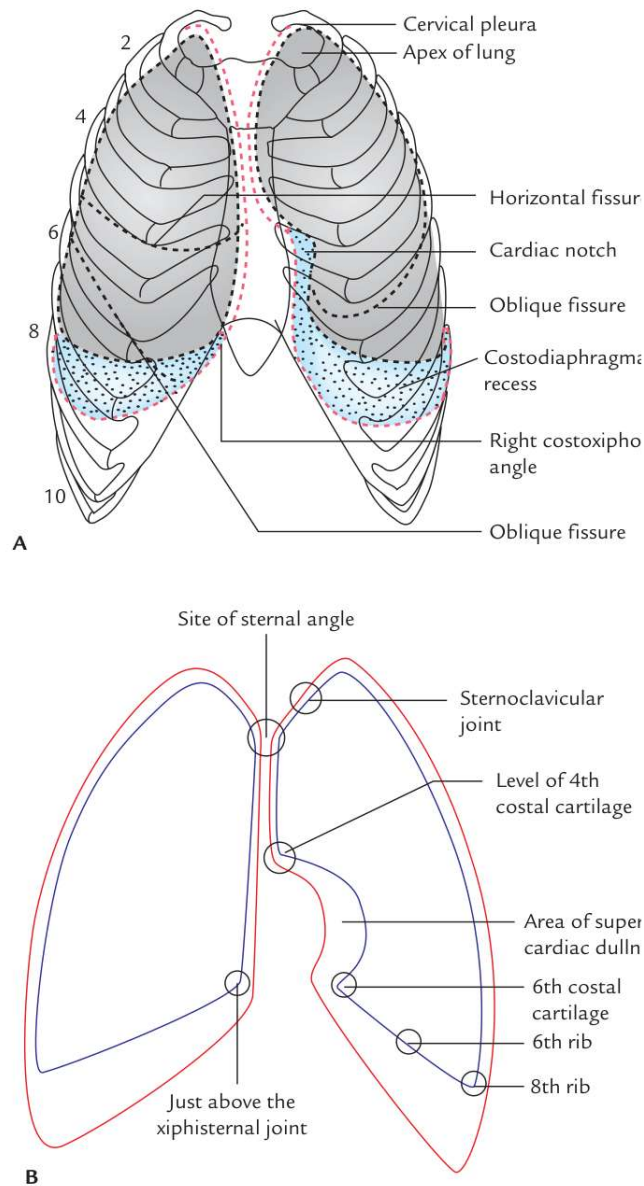


This is the most useful clinical image for auscultation. Anteriorly, upper lobes dominate; the right middle lobe is best approached from the anterior/lateral chest. Posteriorly, except for the apices, most of the visible lung area corresponds to the lower lobes.

Auscultation guide

- Right superior lobe: audible above the 4th rib.
- Right middle lobe: audible between the 4th and 6th ribs.
- Lower lobes of both lungs: audible below the 6th rib anteriorly and best examined posteriorly, especially near the triangle of auscultation.
- Always visualize the oblique fissure: it runs from near T5 posteriorly to the 6th costal cartilage anteriorly.

SURFACE MARKINGS

Fig. 18.9 - Surface markings of lung and pleura

Blue line indicates lung outline and red line indicates pleural outline. The lower border of lung lies about two rib spaces above the lower border of pleura: 6th rib in midclavicular line, 8th rib in midaxillary line and 10th rib near lateral border of erector spinae.

High-yield surface levels

- Lower border of lung: 6th rib at midclavicular line, 8th rib at midaxillary line, 10th rib near lateral border of erector spinae; ends about 2 cm lateral to T10 spine.
- Left anterior border deviates laterally behind 4th and 5th intercostal spaces to form cardiac notch.
- Oblique fissure: line from 2.5 cm lateral to T5 spine to 6th costal cartilage about 4 cm from midline.
- Horizontal fissure: line along 4th costal cartilage to meet the oblique fissure where it crosses 5th rib.

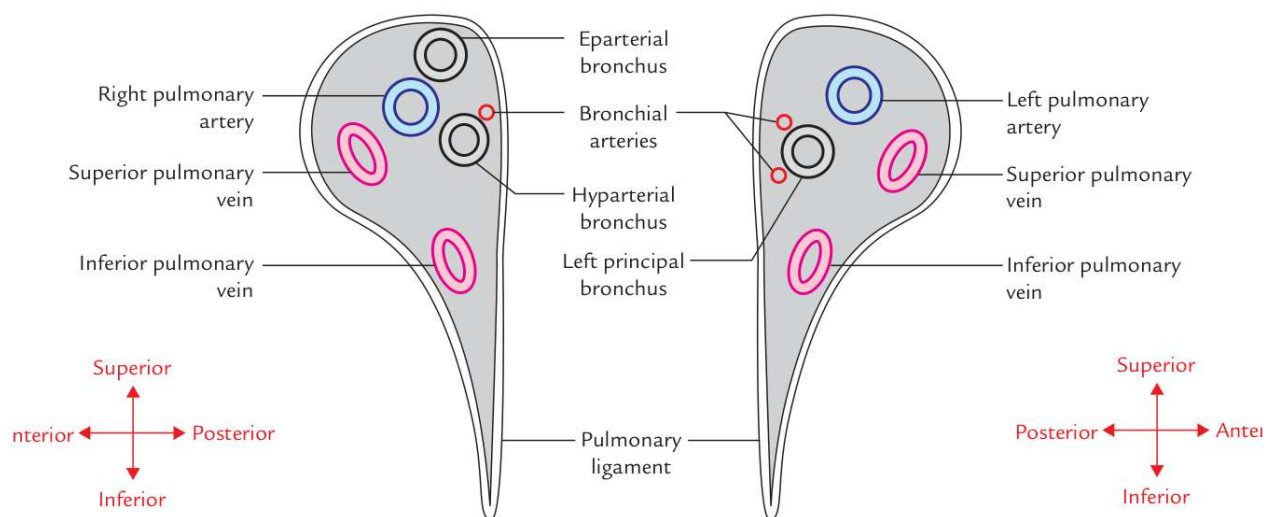
ROOT AND HILUM OF LUNG

Definition and components

The root of lung is a short broad pedicle connecting the medial surface of the lung to the mediastinum. The hilum is the area on the mediastinal surface through which structures enter and leave the lung.

- Components include bronchus/bronchi, pulmonary artery, two pulmonary veins, bronchial arteries, bronchial veins, lymphatics and anterior/posterior pulmonary plexuses.
- The root of lung lies opposite the bodies of T5, T6 and T7 vertebrae.
- The root is enclosed in a tubular sheath derived from mediastinal pleura.

Fig. 18.8 - Arrangement of structures in right and left lung roots



From anterior to posterior, the order is V-A-B: superior pulmonary Vein, pulmonary Artery and Bronchus. From above downward, right side differs because the eparterial bronchus lies above the pulmonary artery, while on the left the pulmonary artery is superior to the left principal bronchus.

Aspect	Relations of root of lung
Anterior	Phrenic nerve, anterior pulmonary plexus; superior vena cava on right side only
Posterior	Vagus nerve, posterior pulmonary plexus; descending thoracic aorta on left side only
Superior	Arch of azygos vein on right side; arch of aorta on left side
Inferior	Pulmonary ligament

Clinical correlation - hilar shadow

In a PA chest radiograph, each lung root casts a radiopaque hilar shadow in the medial one-third of the lung field. The shadow is mainly due to pulmonary vessels seen end-on. Enlargement of bronchopulmonary lymph nodes increases the density of the hilar shadow.

Root and hilum revision box

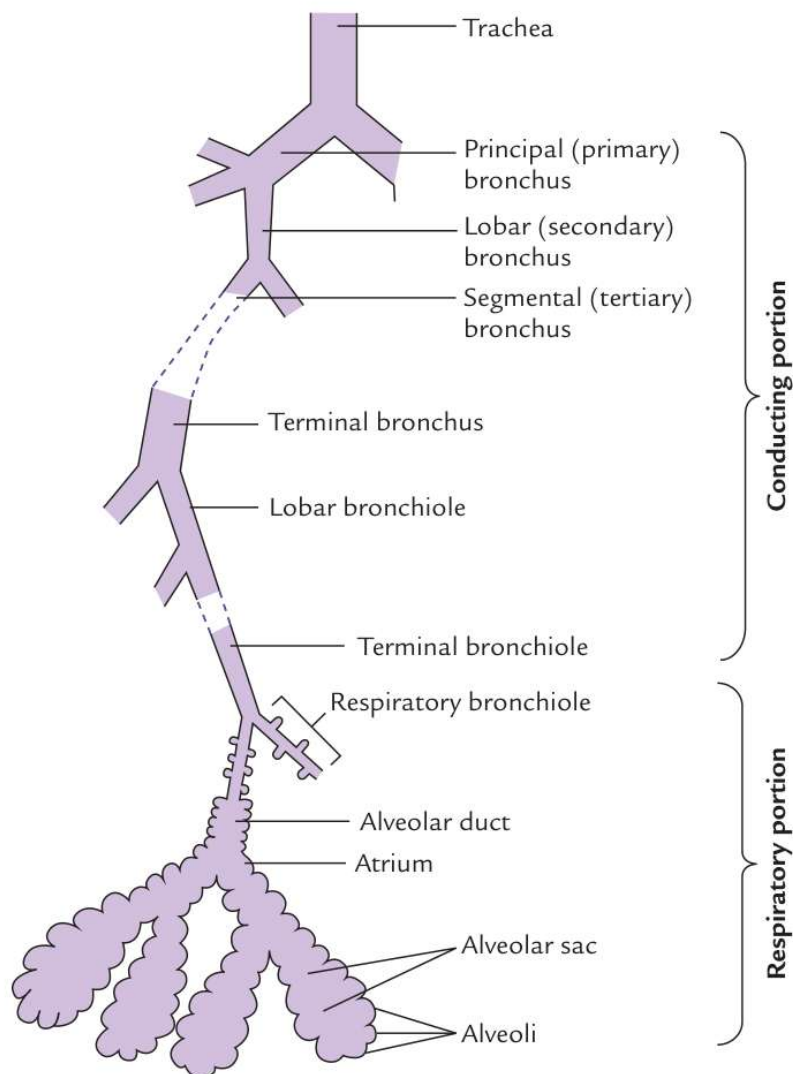
- Hilum is the doorway on the medial surface; root is the pedicle that contains structures passing through the hilum.
- Anterior-to-posterior order is the most stable arrangement: superior pulmonary Vein, pulmonary Artery, Bronchus = VAB.
- Right side has an eparterial bronchus above the pulmonary artery and a hyparterial bronchus below it.
- Left side has one principal bronchus at hilum, placed below the left pulmonary artery.
- Pulmonary ligament hangs inferior to the root and allows movement of root structures during respiration.

Reading the hilum in practical diagrams

In practical spotters, first decide the side of the lung using the cardiac notch, lingula, number of bronchi at the hilum and the arrangement of pulmonary artery. Then identify the pulmonary veins: they are usually placed more anterior and inferior, and they are coloured red in many diagrams because they carry oxygenated blood. The bronchus is usually the more posterior tubular structure. This method is safer than using lobes alone because fissures and lobes may vary.

Hilum comparison	Right lung	Left lung
Uppermost structure	Eparterial bronchus	Pulmonary artery
Bronchi at hilum	Two: eparterial and hyparterial	One: left principal bronchus
Superior relation	Arch of azygos vein	Arch of aorta

BRONCHIAL TREE AND PRINCIPAL BRONCHI

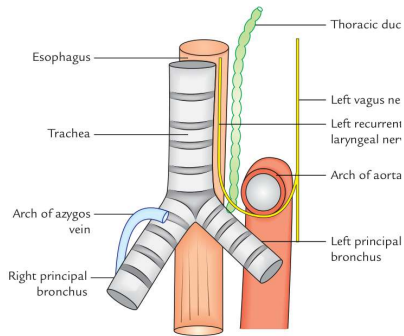
Fig. 18.11 - Bronchial tree**Fig. 18.11** Bronchial tree.

The conducting portion carries air from trachea to terminal bronchioles. The respiratory portion begins near respiratory bronchioles and includes alveolar ducts, atria, alveolar sacs and alveoli, where gas exchange occurs.

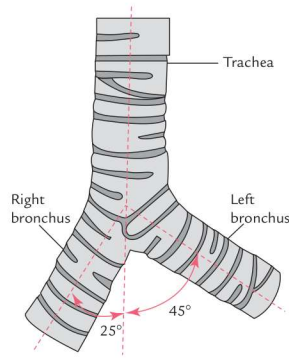
Branching pattern

Trachea -> principal bronchus -> lobar bronchus -> segmental bronchus -> terminal bronchiole -> respiratory bronchiole -> alveolar duct -> atrium -> alveolar sac -> alveoli.

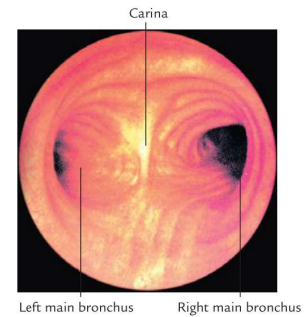
- Right principal bronchus gives three lobar bronchi for three lobes.
- Left principal bronchus gives two lobar bronchi for two lobes.
- Each lobar bronchus divides into segmental bronchi, one for each bronchopulmonary segment.

Fig. 18.12**Fig. 18.12** Relation of principal bronchi.

Relations of principal bronchi: the left bronchus passes below the arch of aorta and in front of esophagus.

Fig. 18.13**Fig. 18.13** Trachea and principal bronch

Right bronchus is more vertical and deviates about 25 degrees; left deviates about 45 degrees.

Fig. 18.14**18.14** Lower end of trachea and its main branch on bronchoscopy. (Source: Fig. 3.48A, Page 7's Anatomy for Students, Richard L Drake, Wayne

Bronchoscopy view: carina is the sensitive ridge at tracheal bifurcation and is important for cough reflex.

Clinical correlations of bronchi

- Aspirated foreign bodies commonly enter the right principal bronchus because it is shorter, wider and more vertical.
- Bronchoscopy visualizes trachea and bronchi; widening/distortion of the carinal angle suggests tracheobronchial lymph node involvement.
- Bronchogenic carcinoma commonly arises from bronchial epithelium, especially in chronic smokers, and may spread to brain through arterial or venous routes.

PULMONARY UNITS, BLOOD SUPPLY AND VENOUS DRAINAGE

Pulmonary unit

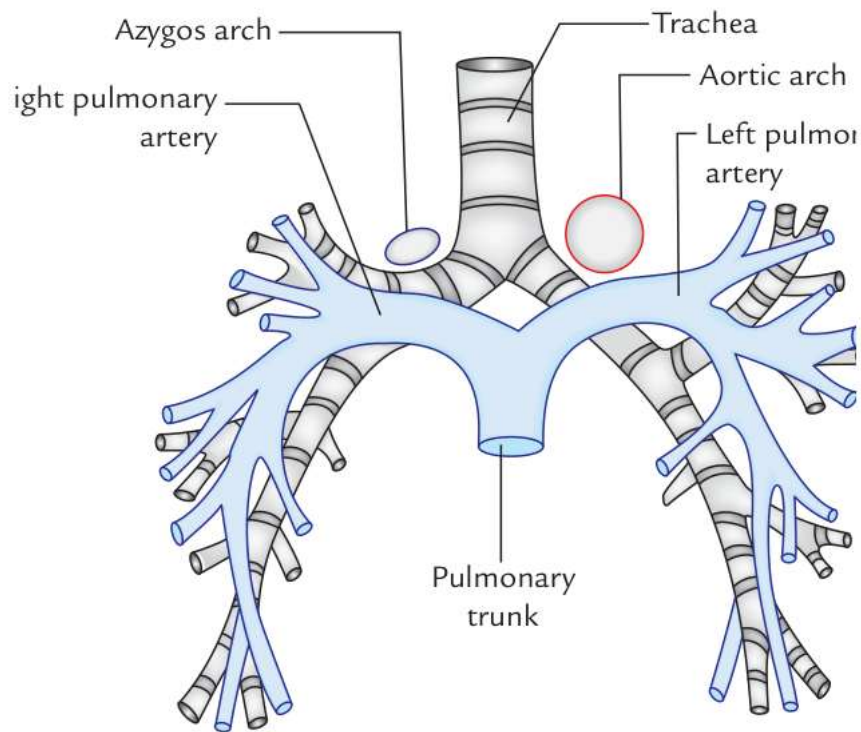
Each pulmonary unit consists of alveolar ducts, atria, air saccules and alveoli.

- Respiratory bronchiole is the transition between conducting and respiratory portions.
- Alveoli are the main sites of gas exchange between inspired air and blood.
- Emphysema damages alveoli and may produce a barrel-shaped chest.

Two arterial systems

Bronchial arteries nourish the bronchial tree and pulmonary tissue.

- Right lung: usually one bronchial artery.
- Left lung: two bronchial arteries from descending thoracic aorta.
- Pulmonary arteries carry deoxygenated blood to the lungs and divide into lobar and segmental branches.

Fig. 18.15 - Relationship of pulmonary arteries to bronchi

The pulmonary arteries lie anterior to the principal bronchi as they enter the hilum. The right pulmonary artery is crossed by the arch of azygos vein, while the left pulmonary artery lies below the aortic arch at about T5 level.

Vessel set	Function and drainage
Bronchial arteries	Nutritive supply to bronchial tree and non-respiratory pulmonary tissue.
Pulmonary arteries	Carry deoxygenated blood from pulmonary trunk to lungs; segmental branches follow bronchial tree.
Bronchial veins	Drain deoxygenated blood from bronchial tree and pulmonary tissue; right to azygos, left to hemiazygos or left superior intercostal vein.
Pulmonary veins	Drain oxygenated blood from lungs to left atrium; their tributaries are intersegmental rather than segmental.

Exam Note

Remember the exception: pulmonary arteries carry deoxygenated blood and pulmonary veins carry oxygenated blood. Also, pulmonary veins run in intersegmental planes, which is important during segmental surgery.

LYMPHATIC DRAINAGE AND NERVE SUPPLY

Fig. 18.16 - Lymphatic drainage of the lungs

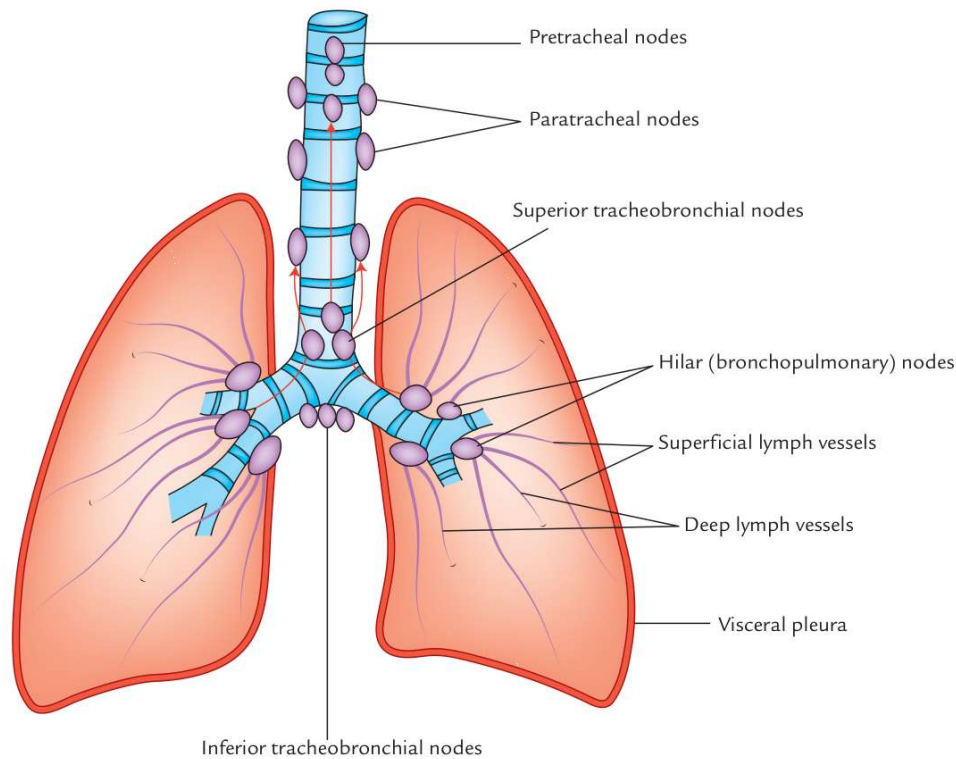


Fig. 18.16 Lymphatic drainage of the lungs.

Superficial lymph vessels drain subpleural tissue; deep lymph vessels drain bronchi, pulmonary vessels and connective tissue septa. Both ultimately drain into bronchopulmonary (hilar) nodes, then tracheobronchial, pretracheal and paratracheal nodes.

Lymph flow

Subpleural and deep plexuses -> bronchopulmonary/hilar nodes -> superior and inferior tracheobronchial nodes -> pretracheal and paratracheal nodes -> bronchomediastinal trunks.

- Right side finally drains to right lymphatic duct.
- Left side finally drains to thoracic duct.
- Clinically important because lung cancer spreads through lymphatics.

Nerve supply

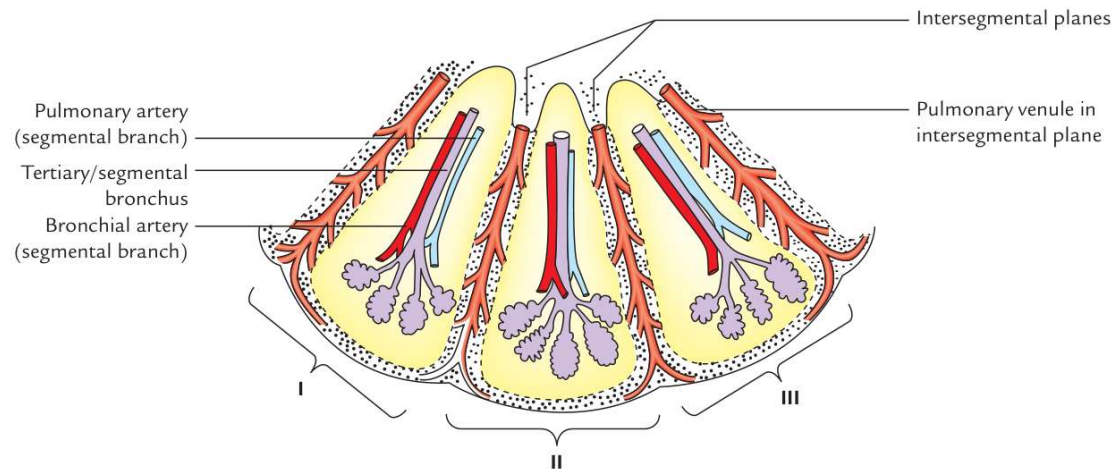
Parasympathetic fibres arise from vagus nerve; sympathetic fibres arise from T2-T5 spinal segments.

- Parasympathetic: bronchoconstriction, vasodilatation and increased mucus secretion.
- Sympathetic: bronchodilatation, vasoconstriction and decreased mucus secretion.
- Afferents from bronchial mucosa and alveolar stretch receptors travel through both systems.

BRONCHOPULMONARY SEGMENTS

Definition

Bronchopulmonary segments are well-defined, wedge-shaped sectors of the lung aerated by tertiary or segmental bronchi. They are anatomical, functional and surgical units of the lungs.

Fig. 18.17 - Schematic bronchopulmonary segments

17 Schematic diagram showing three bronchopulmonary segments (I, II, and III).

Each segment receives a segmental bronchus and a segmental branch of pulmonary artery. The pulmonary venules lie in intersegmental connective tissue planes, forming important landmarks during segmental resection.

Characteristic features

- Subdivision of a lobe of lung.
- Pyramidal in shape: apex points toward hilum, base points toward lung surface.
- Surrounded by connective tissue.
- Aerated by a segmental bronchus.
- Each segment has its own segmental branch of pulmonary artery.
- Each segment has its own lymphatic drainage and autonomic supply.
- Segmental veins run in intersegmental planes, not inside the segment.

Lung	Lobe	Segments
Right	Superior	1 Apical; 2 Posterior; 3 Anterior
Right	Middle	4 Lateral; 5 Medial
Right	Inferior	6 Superior/apical; 7 Medial basal; 8 Anterior basal; 9 Lateral basal; 10 Posterior basal
Left	Superior	1 Apical; 2 Posterior; 3 Anterior; 4 Superior lingular; 5 Inferior lingular
Left	Inferior	6 Superior/apical; 7 Medial basal; 8 Anterior basal; 9 Lateral basal; 10 Posterior basal

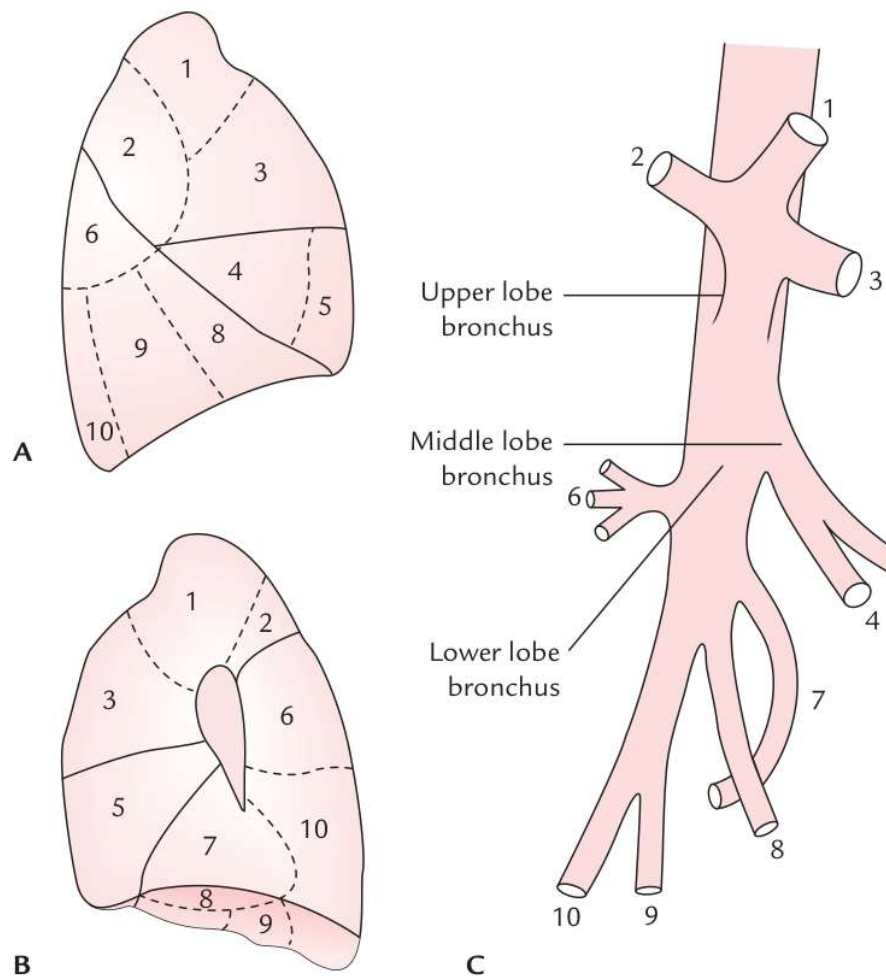
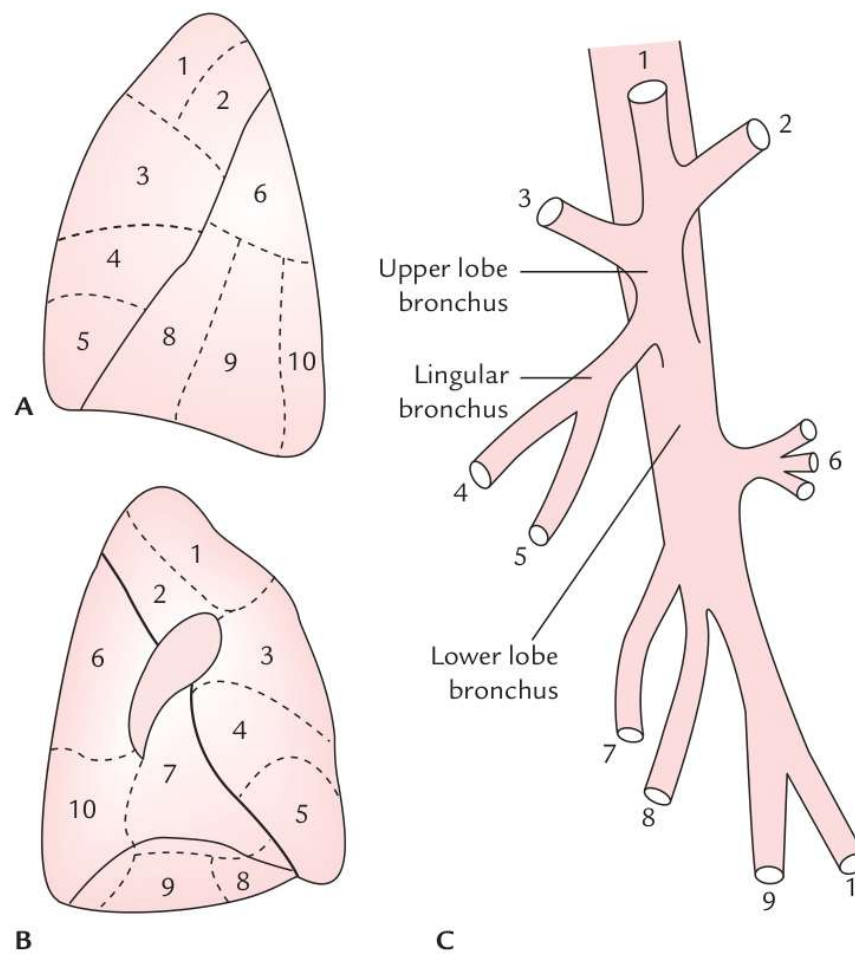
Fig. 18.18 - Bronchopulmonary segments of right lung

Fig. 18.18 Bronchopulmonary segments of the right lung as seen on: **A**, lateral aspect; **B**, medial aspect; **C**, lobar and segmental bronchi.

Right lung has 10 segments: three superior, two middle and five inferior. The middle lobe contributes lateral and medial segments. Basal segments of lower lobe are important in bronchiectasis.

Right-segment map before comparing with left lung

- Right upper lobe: 1 apical, 2 posterior, 3 anterior.
- Right middle lobe: 4 lateral, 5 medial.
- Right lower lobe: 6 superior/apical, 7 medial basal, 8 anterior basal, 9 lateral basal, 10 posterior basal.
- When comparing both lungs, remember that the left superior lobe replaces the right middle lobe region with superior and inferior lingular segments.

Fig. 18.19 - Bronchopulmonary segments of left lung**Fig. 18.19** Bronchopulmonary segments of the left lung seen on: **A**, lateral aspect; **B**, medial aspect; **C**, lobar ar

Left lung has 10 named segments in this classification. The superior lobe includes apical, posterior, anterior, superior lingular and inferior lingular segments; the inferior lobe has five lower-lobe segments.

Clinical value - segmental resection

Knowledge of bronchopulmonary segments allows conservative lung surgery. Because each segment has its own bronchovascular supply and potential intersegmental planes, a diseased segment affected by localized tuberculosis, bronchiectasis or benign neoplasm can be removed while surrounding lung tissue is preserved. During dissection, intersegmental veins should not be ligated unnecessarily because they drain adjacent healthy segments.

APPLIED ANATOMY AND RAPID REVISION

Condition / procedure	Anatomical basis
Pancoast syndrome	Apical lung carcinoma may involve T1 nerve root, sympathetic chain, first rib, subclavian vessels or phrenic nerve.
Foreign body aspiration	Right principal bronchus is shorter, wider and more vertical, so aspirated objects commonly enter right lung.
Bronchial asthma	Bronchospasm narrows bronchioles; expiration is especially difficult and mucus clearance is impaired.
Bronchiectasis	Permanent dilatation of bronchi/bronchioles; basal segments of lower lobes are prone.
Aspiration pneumonia	In supine position, material commonly enters superior/apical segment of lower lobe, especially on right side.
Bronchoscopy	Carina is a key landmark and sensitive cough-reflex area; distortion of carina can indicate tracheobronchial node disease.
Lung cancer lymph spread	Lymph drains through bronchopulmonary/hilar nodes and then tracheobronchial/pretracheal/paratracheal nodes.
Segmental resection	Segmental bronchus and pulmonary artery branch supply a segment; veins lie between segments.

One-page memory list

- Right lung: 3 lobes, 2 fissures, 10 segments; larger, shorter, broader.
- Left lung: 2 lobes, 1 fissure, cardiac notch, lingula, 10 named segments in this classification.
- Root anterior-to-posterior: V-A-B = vein, artery, bronchus.
- Lower border of lung: 6th rib MCL, 8th rib MAL, 10th rib near erector spinae.
- Right principal bronchus: shorter, wider, more vertical - common path for foreign bodies.
- Bronchopulmonary segment = anatomical + functional + surgical unit.
- Segmental veins are intersegmental - protect them during segmental resection.

QUICK SELF-TEST

Question	Recall answer
Which bronchus receives aspirated foreign body most often?	Right principal bronchus - shorter, wider and more vertical.
What is VAB in root of lung?	Anterior-to-posterior order: Vein, Artery, Bronchus.
Which lung has lingula?	Left lung; it is a projection of the superior lobe.
Which nodes are first important in lung lymph drainage?	Bronchopulmonary or hilar lymph nodes.
Where do pulmonary veins run in segments?	Intersegmental planes.
Which segment is commonly involved in aspiration in supine position?	Superior/apical segment of lower lobe, especially on right side.
What makes a segment surgical?	Independent segmental bronchus and artery with connective tissue planes between segments.

What is the lower border of lung at midaxillary line?	8th rib.
Which fibres cause bronchodilatation?	Sympathetic fibres.
Which fibres increase mucus secretion?	Parasympathetic fibres.
Which fissure belongs only to right lung?	Horizontal fissure.
Which surface shows hilum?	Medial surface.
Which nerve passes anterior to lung root?	Phrenic nerve.
Which nerve passes posterior to lung root?	Vagus nerve.
What does a segmental bronchus aerate?	One bronchopulmonary segment.
Which arteries nourish bronchial tree?	Bronchial arteries.

END OF NOTES

Revise diagrams first, then memorize tables and clinical correlations.